

- Feb 2023 **Lunar Gravitational-Wave Antenna - white paper kick-off**, Gran Sasso Science Institute, L'Aquila (IT).
Session chair: "GW science and multi-messenger astronomy" (co-chair with R. Serafinelli; led discussion)
- Nov 2021 **Neutron Star Astrophysics at the Crossroads: Magnetars and the Multimessenger Revolution (IAU Symposium 363)**, Gran Sasso Science Institute, L'Aquila (IT) (online).
Session chair: "NS through GW observations: modelling, search and implications in fundamental physics and astrophysics"
- Nov 2021 **Neutron Star Astrophysics at the Crossroads: Magnetars and the Multimessenger Revolution (IAU Symposium 363)**, Gran Sasso Science Institute, L'Aquila (IT) (online).
Poster session chair
- May 2019 **Physics & Astrophysics at the Extreme (PAX)**, EGO, Pisa (IT).
Session chair: "Compact Binary Coalescences" (co-chair with S. Vitale, J. Veitch and T. Dent)
- Aug 2017 **eXtreme Matter meets eXtreme Gravity**, Montana State University, Bozeman (USA).
Session chair: "New directions and exotic scenarios" (co-chair with Tanja Hinderer)

Invited seminars and talks

- Jun 2027 **[I71] TBA**, Invited conference talk: *Synergies for Black-Hole Binaries: from perturbative techniques to strong-field gravity*, Galileo Galilei Institute, Florence (IT).
- Aug 2026 **[I70] Matters of gravity: the interplay between environmental and beyond-GR effects in asymmetric binaries**, Invited conference talk: *New Horizons for Psi*, Beijing (CHN).
- Jun 2026 **[I69] How does beyond vacuum General Relativity sound?**, Invited conference talk: *Symphony of Spacetime: Celebrating the Discovery of Black Hole Quasinormal Modes* by C. V. Vishveshwara, Raman Research Institute, Bangalore (IN).
- Feb 2026 **[I68] Challenging the vacuum binary paradigm with gravitational wave observations**, Invited workshop talk: *Gravity Workshop: Mergers and Beyond*, University of Tübingen, Tübingen (DE).
- Feb 2026 **[I67] Probing Vacuum General Relativity: Where to Look and What to Seek?**, Invited seminar: *Gr@v Seminar*, University of Aveiro, Aveiro (PT).
- Dec 2025 **[I66] Dancing in the Dark (Matter): How Astrophysical Environments Shape Black Hole Encounters**, Invited seminar, Italian Academy of Advanced Studies, Columbia University, New York (USA).
- Dec 2025 **[I65] Scale Hierarchies in Gravitational-Wave Binaries: A New Path for Probing Fundamental Physics**, Invited seminar: *Theory Seminar*, Columbia University, New York (USA).
- Oct 2025 **[I64] Black hole spectroscopy beyond Kerr with ParSpec**, Invited working group talk: *Beyond GR & Ringdown Group – LISA Fundamental Physics Working Group*, (online).
- Oct 2025 **[I63] Scale Asymmetries in Gravitational Wave Sources and Their Physics Payoff**, Invited conference talk: *10 Years of Gravitational Wave Discovery: Advancing the Frontiers of Gravitational Physics*, Indian Institute of Technology Gandhinagar, Gandhinagar (IN).
- Jul 2025 **[I62] Environmental effects on black hole physics**, Invited conference talk: *GR24 & Amaldi 16*, Glasgow (GB).
- Jun 2025 **[I61] Asymmetric binaries: a tale of two sides**, Invited conference talk: *Black Holes, Neutron Stars, and Gravitational Waves @ Black Sea*, Varna (BG).
- May 2025 **[I60] Matters of Relativity**, Invited workshop talk: *Gravitational Wave Probes of Black Hole Environments*, IFPU, Trieste (IT).
- Mar 2025 **[I59] Probing Fundamental Physics Through Asymmetric Binary Mergers**, Invited seminar, University of Perugia, Perugia (IT).

- Mar 2025 **[158] Gravitational wave observations of black hole mimickers: where do we look, and what do we look for**, Invited conference talk: *Black Hole Mimickers: From Theory to Observation*, Princeton University, Princeton (USA).
- Oct 2024 **[157] Matter of Love**, Invited colloquium, INAF-Astronomical Observatory of Abruzzo, Teramo (IT).
- Sep 2024 **[156] Detecting fundamental fields with LISA observations of extreme mass ratio inspirals**, Invited conference talk: *9th Roma International Conference on Astro-Particle Physics*, Frascati (IT).
- Jul 2024 **[155] EMRIs: Beyond-GR, Ringdown and Environmental Effects**, Invited conference talk: *15th LISA Symposium*, Dublin (IE).
- Jul 2024 **[154] Dark matter signatures around dirty black hole binaries**, Invited conference talk: *New Horizons for Psi*, Instituto Superior Técnico, Lisbon (PT).
- Jun 2024 **[153] Fundamental fields, scalar charges and Extreme Mass Ratio Inspirals**, Invited conference talk: *Athens Symposium on Exploring the Universe*, National Hellenic Research Foundation, Athens (EL).
- Mar 2024 **[152] Gravitational wave signatures of asymmetric binaries surrounded by generic dark matter distributions**, Invited journal club talk: *Cosmology Journal Club*, University of Padua, Padua (IT) (online).
- Feb 2024 **[151] How dirty can a black hole be**, Invited workshop talk: *The New Era of Multi-Messenger Astroparticle Physics*, IFPU, Trieste (IT).
- Feb 2024 **[150] A moon window onto deciHz gravitational wave sources**, Invited seminar, Institute of Space and Astronautical Science, Tokyo (JPN).
- Oct 2023 **[149] Black hole spectroscopy beyond Kerr**, Invited conference talk: *GravityShapePisa*, University of Pisa, Pisa (IT).
- Aug 2023 **[148] The relevance of being small: detecting fundamental fields with LISA observations of Extreme Mass Ratio Inspirals**, Invited seminar (webinar), Latin American Webinar on Physics, (online).
- Aug 2023 **[147] Fundamental fields, asymmetric binaries, and LISA observations within a self-force approach**, Invited conference talk: *Gravitational Waves Meet Effective Field Theories*, Benasque (ES).
- Aug 2023 **[146] Astrophysical and Waveform Systematics from the Working Package Perspective**, Invited conference talk: *Fundamental Physics Working Group Meeting*, Niels Bohr Institute, Copenhagen (DK).
- Jun 2023 **[145] Challenging GR with future detectors: the case of 3G ground based interferometers**, Invited conference talk: *Connecting the Dots*, Max Planck Institute for Gravitational Physics, Potsdam (DE).
- Jun 2023 **[144] Black hole spectroscopy beyond Kerr: agnostic vs theory-based tests with 3G interferometers**, Invited conference talk: *Gravitational Universe: Challenges and Opportunities*, SISSA, Trieste (IT).
- May 2023 **[143] Asymmetric binaries as astrophysical laboratories to probe new fundamental fields**, Invited seminar, University of Aveiro, Aveiro (PT).
- May 2023 **[142] Dirty black hole binaries as probe of dark matter halos**, Invited conference talk: *The Dark Matter and Stars: Multi-Messenger Probes of Dark Matter and Modified Gravity*, Instituto Superior Técnico, Lisbon (PT).
- Apr 2023 **[141] Gravitational waves from extreme-mass-ratio systems in astrophysical environments**, Invited seminar, University of Bremen, Bremen (DE) (online).

- Apr 2023 **[140] Gravitational Wave Sources in the deciHz band**, Invited conference talk: *Lunar Gravitational-Wave Detection*, ICTS, Bangalore (IN).
- Mar 2023 **[139] Asymmetric binaries as probe of fundamental astrophysics**, Invited seminar, Max Planck Institute for Gravitational Physics, Potsdam (DE).
- Nov 2022 **[138] The many faces of Extreme Mass Ratio Inspirals**, Invited seminar, University of Pennsylvania, Philadelphia (USA).
- Jun 2022 **[137] The relevance of being small: testing gravity with LISA and extreme mass ratio inspirals**, Invited seminar, University of Milano-Bicocca, Milan (IT).
- Jun 2022 **[136] Common Tools for the Einstein Telescope Community**, Invited conference talk: *XII Einstein Telescope Symposium*, Hungarian Academy of Sciences, Budapest (HU).
- Apr 2022 **[135] The Science Case of the Lunar Gravitational Space Wave Antenna**, Invited working group talk: *LGWA WG1 – GW Science and Multi-messenger Astronomy*, (online).
- Feb 2022 **[134] Testing General Relativity with LISA observations of extreme mass ratio inspirals**, Invited seminar, Institute of Cosmology and Gravitation, Portsmouth (GB) (online).
- Sep 2021 **[133] Detecting new fundamental fields with LISA**, Invited seminar, Instituto Superior Técnico, Lisbon (PT).
- Mar 2021 **[132] Scalar charges, extreme mass ratio inspirals and LISA**, Invited seminar, LUTH – Observatoire de Paris, Paris (FR) (online).
- Feb 2021 **[131] EMRIs in beyond-GR theories with scalar fields**, Invited working group talk: *LISA EMRI Work Package Meeting*, (online).
- Feb 2021 **[130] Constraining the neutron star equation of state with electromagnetic observations and gravitational waves**, Invited seminar, Brera Astronomical Observatory, Merate (IT) (online).
- Dec 2020 **[129] Extreme mass ratio inspirals as probes of scalar fields**, Invited seminar, University of Tübingen, Tübingen (DE) (online).
- Nov 2020 **[128] Extreme mass ratio inspirals as probes of scalar fields**, Invited seminar, SISSA, Trieste (IT) (online).
- Aug 2020 **[127] Neutron stars in and beyond GR**, Invited conference talk: *Probing Effective Theories of Gravity in Strong Fields and Cosmology*, Kavli Institute for Theoretical Physics, Santa Barbara (USA) (online).
- Jun 2020 **[126] NS EoS using multiband independent measurements of radii and tidal deformabilities**, Invited conference talk: *Science case study and scientific simulations for eXTP*, (online).
- Jun 2020 **[125] Testing alternative theories of gravity with QPO and Fe line**, Invited conference talk: *Science case study and scientific simulations for eXTP*, (online).
- Dec 2019 **[124] The Power of Love**, Invited seminar, University of Ferrara, Ferrara (IT).
- Jun 2019 **[123] Testing the black hole paradigm with gravitational wave binary events**, Invited seminar, Max Planck Institute for Gravitational Physics, Potsdam (DE).
- Feb 2019 **[122] A heated Love for ECO**, Invited seminar, Johns Hopkins University, Baltimore (USA).
- Jun 2018 **[121] Gravitational wave phenomenology of exotic compact objects**, Invited seminar, Albert Einstein Institute, Hannover (DE).
- Jun 2018 **[120] Beyond black holes (exotic compact objects)**, Invited conference talk: *Numerical Relativity Beyond General Relativity*, Benasque (ES).
- Mar 2018 **[119] Solving the relativistic inverse stellar problem through gravitational-wave observations of binary neutron stars**, Invited conference talk: *Problemi Attuali di Fisica Teorica 2018*, Vietri sul Mare (IT).

- Mar 2018 **[I18] Staring at the black hole horizon with gravitational wave observations**, Invited seminar, Kindai University, Osaka (JPN).
- Jan 2018 **[I17] Gravitational waves as a probe of the equation of state of nuclear matter**, Invited seminar, Sapienza University of Rome, Rome (IT).
- Jan 2018 **[I16] Observational signatures of exotic compact objects**, Invited seminar, Instituto de Astrofísica e Ciências do Espaço, Lisbon (PT).
- Dec 2017 **[I15] Challenging the black hole scenario with gravitational wave observations**, Invited seminar, University of Nottingham, Nottingham (GB).
- Oct 2017 **[I14] The strong gravity regime in the gravitational wave era**, Invited seminar, SISSA, Trieste (IT).
- May 2017 **[I13] Strong-field tests of General Relativity in the black hole arena**, Invited seminar, Aristotle University of Thessaloniki, Thessaloniki (EL).
- Mar 2017 **[I12] Signals from the horizon: General Relativity and the strong-field gravity regime**, Invited seminar, Goethe University Frankfurt, Frankfurt (DE).
- Sep 2016 **[I11] Constraining modified theories of gravity using the stochastic gravitational-wave background**, Invited conference talk: *Gravitational Waves in Modified Gravity Theories*, National Technical University of Athens, Athens (EL).
- Aug 2016 **[I10] Fab Four Neutron Stars**, Invited conference talk: *First TUSUR Cosmological Conference*, Tomsk State University of Control and Radioelectronics, Tomsk (RUS).
- May 2016 **[I9] Gravitational waves: a new window onto the high-energy Universe**, Invited working group talk: *Astronomy Group Meeting*, University of Stuttgart, Stuttgart (DE).
- Mar 2016 **[I8] Relativistic tidal effects in non-standard Kerr spacetime**, Invited conference talk: *Problemi Attuali di Fisica Teorica 2016*, Vietri sul Mare (IT).
- Jun 2015 **[I7] Quasi-periodic oscillations of accreting black holes and the strong-field gravity regime**, Invited seminar, Instituto de Ciencias Físicas, UNAM, Mexico City (MEX).
- Dec 2014 **[I6] Gravitational waves and the physics of compact objects: from gamma-ray bursts to I-Love-Q**, Invited seminar, University of Melbourne, Melbourne (AUS) (online).
- Sep 2014 **[I5] Gravitational waves and the physics of compact objects: from gamma-ray bursts to I-Love-Q**, Invited seminar, Center for Relativistic Astrophysics, Georgia Institute of Technology, Atlanta (USA).
- Apr 2014 **[I4] Low-latency search for gravitational waves from BH–NS binaries in coincidence with short gamma-ray bursts**, Invited conference talk: *Problemi Attuali di Fisica Teorica 2014*, Vietri sul Mare (IT).
- Feb 2014 **[I3] Triggering short gamma-ray burst searches with gravitational-wave events**, Invited conference talk: *SFB-Transregio Meeting*, Tübingen (DE).
- Jul 2013 **[I2] Tidal effects in BH–NS and NS–NS binaries**, Invited conference talk: *Stochastic Gravitational Wave Background in the Advanced Detector Era*, Nice (FR).
- Mar 2013 **[I1] I-Love-Q (forever)**, Invited seminar, Institute of Cosmology and Gravitation, Portsmouth (GB).

Contributed talks to conferences and workshops

- Sep 2025 **[C33] Einstein Telescope, deciHz Gravitational Wave Observations on the Moon and Beyond**, Johns Hopkins University, Baltimore (USA).
- Apr 2025 **[C32] The prejudice of Love, Theoretical Horizons in Unraveling Relativity, Astrophysics, and Mergers**, Gran Sasso Science Institute, L'Aquila (IT).
- Apr 2024 **[C31] Love, the easy road, Linking Advances in our Understanding of Theoretical Astrophysics and Relativity to Observations**, University of Milano Bicocca, Milano (IT).

- Jul 2022 **[C30] LISA, EMRIs and new fundamental fields in the gravity sector**, *The 14th International LISA Symposium*, (online).
- Dec 2021 **[C29] Extreme mass ratio inspirals as LISA probes of fundamental fields**, *Rencontre commune des Groupes de travail Formes d'onde et Tests de la relativité générale et théories alternatives du GdR Ondes Gravitationnelles*, Institut de Physique Théorique (IPhT), Paris (FR).
- May 2021 **[C28] Ranking the Love for the neutron star equation of state with third-generation detectors**, *GWADW2021 Gravitational Wave Advanced Detector Workshop*, (online).
- Oct 2020 **[C27] Parametrizing GR deviations within normals modes of rotating black holes with ParSpec**, *TeonGrav meeting*, (online).
- Sep 2020 **[C26] ParSpec: a data-analysis framework for black-hole spectroscopy with multiple events**, *The 13th International LISA Symposium*, (online).
- Jan 2020 **[C25] A parametrized ringdown approach for black-hole spectroscopy of spinning black holes**, *Gravitational Waves, Black Holes and Fundamental Physics*, SISSA, Trieste (IT), [Poster].
- Oct 2019 **[C24] Quasi-periodic oscillations as probe of the neutron star equation of state**, *Future of X-ray Timing*, Amsterdam (NL).
- Jul 2019 **[C23] Gravitational waves in massive gravity: waveforms from extreme-mass-ratio mergers**, *22nd International Conference on General Relativity and Gravitation*, Valencia (ES).
- Feb 2019 **[C22] From multi-band observations to the nuclear matter equation of state**, *EPS Conference on Gravitation 2019*, Sapienza University of Rome, Rome (IT).
- Jan 2019 **[C21] Gravitational waves in massive gravity theories: constraints from extreme-mass-ratio mergers**, *Gravitational Waves, Black Holes and Fundamental Physics*, Athens (EL).
- Sep 2018 **[C20] Exotic compact objects and their gravitational-wave signatures**, *TeonGrav Meeting 2018*, Sapienza University of Rome, Rome (IT).
- Jan 2018 **[C19] To be a black hole or not to be a black hole**, *Gravity@Malta*, University of Malta, Valletta (MT).
- Dec 2017 **[C18] Gravitational wave signatures of black hole mimickers**, *X Black Holes Workshop*, Aveiro University, Aveiro (PT).
- Nov 2017 **[C17] Exotic Compact Objects in the Strong Field Regime**, *Multidisciplinary Centre for Astrophysics (CENTRA)*, Lisbon (PT).
- Jun 2017 **[C16] A Montecarlo approach for the relativistic inverse stellar problem**, *Bridging Nuclear and Gravitational Physics: the Dense Matter Equation of State*, ECT, Trento (IT).
- Feb 2017 **[C15] Testing strong-field gravity with quasi periodic oscillations**, *High Throughput X-ray Astronomy in the eXTP Era conference*, Rome (IT).
- Sep 2016 **[C14] Binary BH gravitational wave background as probe of alternative theories of gravity**, *NEB-17 Recent Developments in Gravity*, Mykonos (EL).
- Oct 2015 **[C13] From macro to micro: universal properties of neutron stars**, *The Modern Physics of Compact Stars and Relativistic Gravity 2015*, Yerevan State University, Yerevan (AM).
- Jul 2015 **[C12] Interacting vacuum-energy in a Post-Friedmann expanding Universe**, *14th Marcel Grossman Congress 2015*, Sapienza University of Rome, Rome (IT).
- Jul 2015 **[C11] Testing the strong field gravity regime with QPOs observations**, *14th Marcel Grossman Congress 2015*, Sapienza University of Rome, Rome (IT).
- Jul 2015 **[C10] Slowly-rotating black hole solutions in Horndeski theory**, *IVth NRHEP Network Meeting*, Sapienza University of Rome, Rome (IT).

- Apr 2015 **[C9] Testing the strong field gravity regime with QPOs astrophysical observations**, *Center for Relativistic Astrophysics weekly meeting*, Atlanta (USA).
- Nov 2014 **[C8] Testing Gravity with QPOs of Black holes: the case of Einstein-Dilaton-Gauss-Bonnet theory**, *5th LOFT Consortium meeting*, Rome (IT).
- Apr 2014 **[C7] Coincidence searches of gravitational waves and short gamma-ray bursts**, *San Cugat Forum on Astrophysics*, San Cugat, Barcelona (ES).
- Feb 2014 **[C6] Universal relations for Neutron Stars: Gravitational Waves and Short Gamma-ray Bursts**, *TeonGrav Meeting 2014*, Sapienza University of Rome, Rome (IT).
- Oct 2013 **[C5] Black hole-neutron star mergers**, *Einstein Telescope Meeting 2013*, Albert Einstein Institute, Hannover (GE).
- Jul 2013 **[C4] I Love Q forever**, *An International Workshop. Neutron stars: Nuclear Physics, Gravitational Waves and Astronomy*, University of Surrey, Guildford (GB).
- Jul 2012 **[C3] Post-Newtonian-Affine tidal interactions**, *13th Marcel Grossman Congress 2012*, Stockholm (SE).
- Apr 2012 **[C2] Post-Newtonian-Affine tidal interactions**, *Problemi Attuali di Fisica Teorica 2012*, Vietri sul Mare (IT).
- Jun 2011 **[C1] Tidal interactions in compact binary systems**, *5th Nuclear Physics School*, Otranto (IT).

Publications

Regular papers

- gr-qc/
2607.09310 **[R81]** Susanna Barsanti, Ollie Burke, **Andrea Maselli**, Thomas Sotirou, Andrew Spiers, Niels Warburton, "*The significance of first post-adiabatic contributions for scalar charge measurements with extreme-mass-ratio inspirals*", 2026.
- gr-qc/
2606.18341 **[R80]** Albert Radulea, Marcelo Rubio, Konstantinos Kritos, **Andrea Maselli**, "*Orbital evolution of asymmetric binaries within accreting environments*", *Phys. Rev. D*, 2026.
- gr-qc/
2605.02633 **[R79]** Simone D'Onofrio, Sayak Datta, **Andrea Maselli**, "*Axial tidal Love numbers of black holes in matter environments*", *Phys. Rev. D*, 2026.
- gr-qc/
2603.10116 **[R78]** Sara Gliorio, Matteo Della Rocca, Susanna Barsanti, Leonardo Gualtieri, **Andrea Maselli**, Thomas Sotirou, "*Adiabatic evolution of asymmetric binaries on generic orbits with new fundamental fields I: characterization of gravitational wave fluxes*", *Phys. Rev. D*, 2026.
- gr-qc/
2512.07959 **[R77]** Matteo Della Rocca, Laura Pezzella, Matteo Emanuele Berti, Leonardo Gualtieri, **Andrea Maselli**, "*Quasinormal ringing of Kerr black holes. III. Excitation coefficients for equatorial inspirals from the innermost stable circular orbit*", *Phys. Rev. D* 114, 044036, 2026.
- gr-qc/
2507.16904 **[R76[†]]** Giorgio Nicolini, **Andrea Maselli**, Miguel Zilhao, "*Dancing in the dark: probing Dark Matter through the dynamics of eccentric binary pulsars*", *SciPost Phys.* 20, 068, 2026.
- gr-qc/
2507.04471 **[R75*]** Sayak Datta, **Andrea Maselli**, "*A multi-parameter expansion for the evolution of asymmetric binaries in astrophysical environments*", *SciPost Phys.* 20, 080, 2026.
- gr-qc/
2503.16649 **[R74[†]]** Sara Gliorio, Emanuele Berti, **Andrea Maselli**, Nicholas Speeney, "*Extreme mass ratio inspirals in dark matter halos: Dynamics and distinguishability of halo models*", *Phys. Rev. D* 112, 124050, 2025.

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